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MILITARY SPECIFICATION

SEMITRAILER, FLATBED: BREAKBULK/CONTAINER TRANSPORTER, 34 TON, M872A4

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercially designed 34 ton payload platform, three-axle, dual purpose (breakbulk/container) semitrailers, M872A4, for use in highway transport of breakbulk cargo and a variety of intermodal cargo containers while being towed by a suitable truck-tractor (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
FEDERAL

V-T-285	- Thread, Polyester.
NN-P-530	- Plywood, Flat Panel.
TT-C-490	- Cleaning Method for Ferrous Surfaces and Pretreatment for Organic Coatings.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-W-3912	- Wood Parts and Wood Substitutes, Fabricated: for Transport Vehicle Bodies and Similar Applications.
MIL-A-8421	- Air Transportability Requirements, General Specification for.
MIL-C-20696	- Cloth, Coated, Polyester or Nylon, Waterproof.
MIL-R-24049	- Rope, Polypropylene.
MIL-P-46105	- Primer Coating; Weld-Through, Zinc-Rich.
MIL-R-46164	- Rustproofing of Military Vehicles and Trailers.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-V-62038	- Vehicles, Wheeled: Preparation for Shipment and Storage of.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attribute.
MIL-STD-193	- Painting Procedures and Markings for Vehicles, Construction Equipment and Materiel Handling Equipment.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-810	- Environmental Tests Methods and Engineering Guidelines.

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MIL-STD-1261	- ARC Welding Procedures for Constructional Steels.
MIL-STD-1366	- Material Tranportation System Dimensional and Weight Constraints, Definition of.
MIL-STD-45662	- Calibration Systems Requirements.
MS 20230	- Grommets, Metallic, Plain and Spur, With Washer, Type I and Type III.
MS 52125	- Composite Light-Tail, Stop, Turn and Marker.
MS 75021	- Connector, Receptacle, Electrical-12 Contact, Intervehicular, 28-Volt, Waterproof.
MS 87006	- Hook, Chain-S.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

7722333	- Grommet Electrical Connector.
7723309	- Nut Grommet-Retaining Size 28.
7731428	- Cover Assembly, Receptacle.
10900880	- Strap, Webbing Universal Tiedown.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations (FMCSR)

393.45	- Brake Tubing and Hose Adequacy.
393.46	- Brake Tubing and Hose Connections.
393.52	- Brake Performance.
393.70	- Coupling Devices and Towing.
393.86	- Rear End Protection.
393.100	- Protection against Shifting or Falling Cargo.
393.106	- Front-End Structure.

Federal Motor Vehicle Safety Standards (FMVSS)

108	- Lamps, Reflective Devices and Associated Equipment.
121	- Air Brake Systems-Trucks, Busses and Trailers.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A36	- Standard Specification for Structural Steel.
ASTM A513	- Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing, Spec. for.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, (SAE)

SAE Standard and Recommended Practices

SAE J267	- Wheels/Rims-Trucks-Test Procedures and Performance Requirements, Recm. Practice.
SAE J534	- Lubrication Fittings.
SAE J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
SAE J682	- Rear Wheel Splash and Stone Throw Protection.
SAE J700	- Fifth Wheel Kingpin - Commerical Trailers and Semitrailers.
SAE J702	- Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer.
SAE J753	- Lubrication Chart - Construction and Industrial Machinery.
SAE J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring, Recm. Practice.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

Yearbook

(Application for copies should be addressed to the Tire and Rim Association Inc., 3200 West Market Street, Akron, OH 44313.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish semitrailers which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent semitrailers delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Life cycle operations. The reliability and maintainability requirements shall be demonstrated over the life cycle operations of the semitrailer when operated for a total of 6,000 miles (mi) per vehicle in accordance with the operational profile in table I, the payload and distribution requirement of 3.2.1, and the environmental conditions of 3.2.2 (see 4.8.3).

TABLE I. Operational profile.

Course	Mileage, percent (%)	Loading Condition	Speed, miles per hour (mph)
Primary road 1/	45	at rated payload	55
	5	empty	55
Improved secondary road 1/	45	at rated payload	30
	5	empty	30

1/ See 6.3.1 and 6.3.2 for definitions.

3.2.1 Payload and distribution. The semitrailer shall be capable of transporting a single American National Standards Insitute/International Organization for Standardization (ANSI/ISO) standard size 40-foot (ft) long container; a combination of two ANSI/ISO 20-ft containers; breakbulk cargo; TRICON and QUADCON containers; and any combination of containers of standard and non-standard lengths up to 40 ft, breakbulk cargo and vehicles. Regardless of the load composition, the semitrailer shall be capable of loads with combined weights no greater than 67,200 pounds (lb) of breakbulk cargo uniformly distributed over the load space, as specified in table II, without permanent deformation to the frame (see 4.8.3.1).

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TABLE II. Payload containers.

Quantity	Length, ft	Weight, lb
1	40 (ISO)	67,200
1	35	54,000
1	27 1/	60,000
1	24 1/	50,000
2 or 1	20 (ISO) 1/	67,200/2 45,000/1
4 or less	10 (ISO)	67,200 total 22,400 each
6 or less	6 2/3 (ISO)	67,200 total 15,000 each
8 or less	5 (ISO)	67,200 total 10,000 each

1/ The container locking devices shall center 20, 24, and 27 ft containers within three ft of the longitudinal midpoint of the platform.

3.2.1.1 Tracked vehicle payload. The semitrailer shall be capable of transporting two M113 tracked vehicle series or any two vehicle based upon the M113 chassis. The load of each tracked vehicle shall be distributed equally over an area of 12 x 115 inches (in), each along the left and right edges of the platform. The tracked vehicles shall be placed on the semitrailer for optimum practical weight distribution (see 4.8.3.1).

3.2.2 Environmental. The semitrailer and components shall withstand and operate under extreme conditions of weather and altitude in ambient air temperatures ranging from minus (-) 25 degrees Fahrenheit (°F) to plus (+) 125°F and shall withstand storage at -80°F without deterioration that may cause failure of any component (see 4.8.3.2).

3.2.3 Reliability. The reliability of the semitrailer shall have a design goal of 6,000 Mean Miles Between Failure (MMBF) to be demonstrated in accordance with 3.2. For calculation of MMBF, a failure shall occur when the semitrailer prevents operation, reduces performance below essential levels, indicates that further operation would be unsafe, or indicates that further operation might result in extensive damage to the equipment. Any of these conditions that cannot be corrected by the crew in one hour (hr) using Basic Issue Items (BII) that are carried on the prime mover or the semitrailer shall be considered a failure. Maintenance and human induced failures are excluded. Securement system reliability shall be demonstrated during test as specified (see 4.8.3.3).

3.2.4 Maintainability. The total maintenance time, excluding driver and crew checks and services shall not exceed 8 manhours during 6,000 mi of specified operation. This equates to a maintenance ratio (MR) of 0.04 at 30 mi of operation equivalent to one hr of operation. The scheduled maintenance

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interval shall be six months or 6,000 mi. Any of these conditions that cannot be corrected by the crew in one hr using BII shall be considered a failure. Maintenance and human induced failures are excluded (see 4.8.3.4).

3.3 Materials. Material used in the manufacture of the vehicle shall be new and of highest commercial quality as normally used for this type of equipment, considering strength, durability, ductility, best engineering practice and the purpose for which the equipment is to be used; and shall be as specified, where applicable (see 4.8.1).

3.3.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials; however, all materials shall be new and unused. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.8.1 and 6.3.3).

3.4 Design and construction. The semitrailer shall be of the heavy-duty commercial container transporter type, modified as necessary to accomodate all containers and breakbulk cargo capabilities specified herein. Except as specified herein, vehicle components, assemblies and accessories to be delivered under contract shall be manufacturer's standard or optional items. The automotive vehicles, components, assemblies and accessories to be procured hereunder shall conform to all FMVSS applicable on the date of manufacture to non-military, non-export semitrailer products, except as otherwise specifid herein (see 4.8.1 and 4.8.2).

3.4.1 Dimensions and clearances. The semitrailer, without payload, uncoupled from the towing vehicle, resting on its landing legs on level ground, and with the platform level, shall conform to the dimensions specified below. The semitrailer shall not exceed the Passe-Partout International (PPI) clearance dimensions as shown in figure 1 (see 4.8.2).

a.	Length of load space	40 ft
b.	Maximum Vehicle width	96 in
c.	Maximum Platform height	55 in
d.	Maximum frame depth at fifth wheel	5-1/2 in
e.	Distance from front of semitrailer to vertical centerline of kingpin	30 $\pm$ 2 in
f.	Upper fifth wheel height, (empty) from the ground to the underside of the upper fifth wheel plate	49 $\pm$ 1 in
g.	Minimum swing clearance from the centerline of the kingpin to any portion of the semitrailer six inches or more below upper fifth wheel plate.	87 in
h.	Maximum swing radius from centerline of kingpin to most distant point on semitrailer nose.	58 in

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3.4.2 Weight. The weight of the semitrailer shall be minimum practicable for service intended and the curb weight shall not exceed 19,179 lb. The Gross Combined Weight (GCW) of the tractor and semitrailer shall be no more than 105,000 lb (see 4.8.2).

3.4.3 Chassis Components. The chassis components shall be adequate to support the payloads specified in 3.2.1 and table II under the operating conditions specified herein (see 4.8.1).

3.4.4 Frame. Main frame members shall be adequately braced with a sufficient number of cross members to prevent permanent deformation. The outside frame members shall run the full length of the load space. The rear cross member shall extend the full width of the semitrailer. With the semitrailer statically loaded and the load distributed as specified in 3.2.1 and table II, the maximum fiber stress in any frame member shall be no greater than one third of the material yield strength. The frame shall be completely electric arc welded (see 4.8.1 and 4.8.2).

3.4.5 Frame splices. Where frame members are spliced, joints shall be reinforced to fulfill the static loading requirements of 3.2.1 and the dynamic loading requirements of 3.4. Frame splices shall be included with technical proposal and shall be designed to minimize stress concentration factors to values below 1.3. All splice welds shall be continuous (see 4.8.1).

3.4.6 Axles and suspension. Axle and suspension system shall have manufacturer's rated capacity at least equal to the load imposed at the ground when the semitrailer is loaded with its specified payload. Suspension clearances shall preclude interference between wheels and with any portion of the semitrailer when operating under all conditions specified. The axles shall be of triaxle configuration of a proven commercial design and shall mount six sets of dual wheels (see 4.8.2).

3.4.6.1 Suspension articulation. Sufficient articulation shall be provided to maintain equal loading on each wheel when semitrailer, with rated payload, is coupled to prime mover with fifth wheel height varying from 50 to 56 in (empty) and under each of the conditions listed below. Differences in height for specified sets of dual wheels shall be measured at center of tread on outer tires (see 4.8.2).

a. With first and third on one side and center wheel on opposite side, elevated four inches above remaining wheels.
b. With any pair of opposite wheels elevated four inches above remaining wheels.
c. With any two pairs of opposite wheels elevated four inches above remaining pair.

3.4.7 Wheels, rims, tires and tubes. All wheels shall be fabricated in accordance with SAE J267. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size tire furnished (see 4.8.1 and 4.8.2).

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3.4.7.1 Tires. Tires shall be radial tube type with highway tread and shall have a rated capacity at least equal to the load imposed on each tire, measured at the ground, with vehicle loaded to specified payload. The tires shall have a minimum size and rating of 10.00 x 20, 14 ply, load range G (see 4.8.1 and 4.8.2).

3.4.7.2 Inner tubes. Inner tubes shall be heavy-duty type and shall be of proper size for tires furnished. Tire flaps shall be provided in accordance with Tire and Rim Association recommendations (see 4.8.2).

3.4.7.3 Tire carrier. One tire carrier shall be provided and installed under the frame at curb side in an accessible location. The means for securing the tire and wheel to the carrier shall be a minimum 1/4 inch proof coil, welded, galvanized, approximately four ft long. It shall be wrapped around the tire and wheel assembly and tire mounting bracket and then securely joined together with a nut, bolt and washers. The chain shall also be attached to the tire carrier with a bolt and lock nut (see 4.8.2).

3.4.7.4 Spare wheel assembly. A spare wheel or rim with inflated tire shall be mounted on the tire carrier specified in 3.4.7.3. The spare tire shall be the same as the tires furnished on the semitrailer (see 4.8.2).

3.4.7.5 Tire wrench. A tire wrench shall be provided with each semitrailer. Stowage provision shall be provided for the wrench and shall adequately secure the wrench to prevent accidental loss (see 4.8.2).

3.4.7.6 Rear wheel splash and stone throw protection. Wheels shall have anti-sail flexible mud flaps at the front and the rear of the wheels. Splash and stone throw protection shall be in accordance with SAE J682 (see 4.8.2).

3.4.7.7 Tire changing capability. The semitrailer shall be provided with a means of raising each tire off the ground so that the wheel and tire assembly may be removed and replaced from either an unloaded or fully loaded semitrailer under all terrain conditions in no more than 30 man/minutes. All tools necessary to remove and replace a wheel and tire assembly, by one soldier, shall be provided. Storage space for the above tools, lockable with a common padlock, shall also be provided (see 4.8.2).

3.4.8 Brakes. Brakes shall be of the full air, internal expanding type and shall conform to FMCSR's 393.45, 393.46 and 393.52. The braking system shall include standard breakaway features, relay emergency valve, air reservoir, manual or automatic slack adjusters, piping, hose connections, glad-hands, spring loaded dust covers or dummy glad-hands equipped with security chains, and all other components required for a complete air-brake system. Air hose fittings and locations shall comform to SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by

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objects striking components. No part of the braking system shall extend below the bottom of wheel rims. Slack adjusters shall be located above the bottom edge of the axle. The performance of the brake system shall conform to FMVSS 121 except for the following (see 4.8.1 and 4.8.2).

a. The semitrailer shall not conform to the "anti-lock" requirements.
b. The semitrailer air reservoir capacity shall be four times the combined volume of all service brake chambers at maximum travel of the pistons or diaphragms instead of the eight specified.
c. The semitrailer shall conform to the "parking brake system" requirements. One set of spring type parking brakes are required on each axle of the tri-axle tandem.

3.4.9 Wheel chock blocks. Four nonwood wheel chock blocks with grab handles shall be provided and installed, with security chain and snap, on the semitrailer. Stowage provisions shall be provided for the chock blocks near the wheels and mounted to the side of the main frame webs (see 4.8.2).

3.4.10 Upper fifth wheel plate. The upper fifth wheel plate shall be adequate for coupling to a full oscillating military fifth wheel, and the fore and aft rocking commercial fifth wheel. It shall be of sufficient size to cover a fifth wheel, 36 inches in diameter, and shall conform to FMCSR 393.70 (see 4.8.2).

3.4.10.1 Kingpin. The upper fifth wheel kingpin shall be 2 inches and shall conform to SAE J700 (see 4.8.2).

3.4.11 Landing gear. The semitrailer shall have two vertical lift, telescopic, nonrotating landing legs, with two speed gears and a handcrank on the curb side. The landing legs shall be equipped with self-leveling skid pads. Supports for the crank extension shafts and clips for holding the crank when folded shall be provided. The landing gear shall support the fully loaded semitrailer and shall withstand, without deformation, the combined static and dynamic forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to preclude the entrance of foreign matter. The landing gear shall have a range of adjustment and rated capacity to vary the height of the upper fifth wheel from 50 inches to not less than 56 inches from the ground with rated payload on the semitrailer. With the semitrailer coupled to a towing tractor and in level position, the clearance under the fully retracted landing gear shall be no less than 14 in (see 4.8.2).

3.4.12 Electrical.

3.4.12.1 Lighting system. The lighting system shall be 12 volts direct current (V dc) with all associated components and shall comply with FMCSR Part 393 and FMVSS 108. The system shall include no less than two composite lights as specified in MS52125-1. Lights and reflectors shall be recessed or

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guarded to protect them from damage during normal operation of the semitrailer. All wiring shall be properly routed and secured to protect against snagging or tearing. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails or semitrailer bumpers. All electrical wiring shall conform to SAE J1292 (see 4.8.1 and 4.8.2).

3.4.12.2 Blackout lighting. The light emitting diode (LED) blackout stop and marker shall operate on 24 V. The emission of any vehicle light source, which may be illuminated in the blackout mode, shall be limited to the visible spectrum, 380 to 700 nanometers. No energy shall be emitted in the 700 to 1200 nanometer portion of the electromagnetic (EM) spectrum. Emission peaks shall not be more than 1% of the peak emission in the visible spectrum. Turn signals and clearance lights are to be inoperative during blackout drive conditions (see 4.8.1).

3.4.12.3 Receptacle system, 12-24 V. The semitrailer shall be equipped with one 12 contact receptacle with cover, and one 7 contact receptacle with cover installed at the front of the semitrailer. The 12 contact receptacle shall conform to MS75021-1 and be equipped with a cover assembly conforming to Drawing 7731428, nut grommet, retainer conforming to Drawing 7723309, and grommet, electrical connector conforming to Drawing 7722333. The 7 contact receptacle shall conform to SAE J560 round socket for an intervehicle connector cable. The 12 contact receptacle shall be provided with necessary components to each circuit to reduce the voltage of a tactical military design truck tractor from 24 V to 12 V shall be connected through the necessary resistors into the 12 V semitrailer circuits as follows:

a. Not used
b. Connect to Circuit Number 3
c. Not used
d. Connect to Circuit Number 1
e. Connect to Circuit Numbers 2 and 6
f. Not used
g. Not used
h. Connect to Citcuit Number 5
i. Not used
j. Not used
k. Not used
l. Not used

In addition, the connection between circuit 2 and 6, on the 7 pin connector, shall have a readily accessible external switch to allow for breaking of this circuit when the 7 pin connector is attached to the towing tractor in lieu of the 12 pin MS connector. Circuits b and j on tactical truck tractors, are combination stop and turn signal indicator circuits. The normal 12 V turn signal lights will function both as turn signals and stop lights and the normal 12 V stop lights will not be in operation when the semitrailer is connected to a towing vehicle with a 24 V power supply. Because of this condition, the number 4 stop light circuit is not connected to the 24 V, 12 contact receptacle (see 4.8.2).

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3.4.13 Rear end protections. The semitrailer rear end protection shall be in accordance with FMCSR 393.86 (see 4.8.2).

3.4.13.1 Dock bumper. A dock bumper, extending not less than 1.5 inches beyond the rear of semitrailer, shall be furnished. The dock bumper shall be full width of vehicle and shall be located approximately at the semitrailer deck level. In addition, two rubber bumpers of good commercial quality suitable for the intended purpose, one at each end of the dock bumper, shall be furnished to provide protection to the rear of the vehicle (see 4.8.2).

3.4.14 Platform. The platform shall have a flat front with square corners and steel stake pockets that are located sufficiently outboard to provide a minimum of 90 inch clear load space width inside the racks for the entire length of the load surface. If protruding type pockets are provided, they shall be protected all around by a steel rub rail with no less than 2 inches wide and 5/16 inch thick. Flooring shall be of hardwood conforming to class 1 of MIL-W-3912 except that laminated hardwood shall not be used. Floorboards shall be no less than 1-3/8 inches thick and of maximum practical length laid lengthwise with tongue-and-groove or shiplap joints. Flooring shall be secured with countersunk cadmium or zinc-plated screws or bolts. No less than two securing screws or bolts shall be installed at each cross sill and two on each side of each butt joint. Butt joints shall be centered over crossmembers and staggered. Top of countersunk screw or bolt head shall be below the top of the wood surface by not less than 1/8 inch and not more than 1/4 in. Flooring shall be flush to 1/8 inch higher than front and rear sills (see 4.8.2).

3.4.14.1 Front panel. Each semitrailer shall be equipped with a combination of front-end structure and front panel. The front panel shall be 48 inches in height above the bed and shall meet the requirements of FMCSR 393.106 The front panel shall be of permanent construction, without openings, and shall allow sufficient clearance at the front for rapid container loading. The front panel shall also incorporate guide ramps near its base to guide the container rearward into its correct position. The front panel shall be hinged in such a manner to reduce the height of the vehicle 12 inches to allow loading. There shall be four hinges and the hinged section of the panel shall fold rearward and in the upright position and shall be locked in position with two lock pins with security chains to prevent loss. The container locks for the front of the front container may be incorporated into the front panel. Each panel shall be finished, primed, and camouflage painted with paint as used on the semitrailer (see 4.8.2).

3.4.14.2 Side and rear racks. The semitrailer shall be furnished with side and rear racks that are located in the stake pocket holes as specified herein. The number of intermediate side sections shall be 18, with intermediate rack section width not exceeding 46 1/2 in. The front forward side and the rear rack sections shall not exceed 50 1/4 and 44 1/2 inches in width respectively. All sections shall be easily removable by one man and show no evidence of binding or jamming. The intermediate side sections shall be interchangeable on both sides, except for the two forward side racks and two rear racks. All sections shall meet and fit so that interfacing channels

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grade.

engage and disengage freely and shall require no use of other than small handtools. The racks shall be constructed using 5/8 inch thick, industrial for exterior grade, plywood panels in accordance with grade B-B of NN-P-530 surfaced on both sides with Resin Impregnated Fiber Sheeting (RIFS) and waterproof edge seal. Finished thickness of the RIFS panel shall be 0.70 inches maximum. The surfacing shall consist of phenolic resin and cellulose fibers with the phenol formaldehyde content no less than 17% by weight. Surfacing shall be bonded to both sides of the plywood panels with waterproof phenolic resin. All edges of the panels shall be sealed with two coats of polyurethane varnish or phenolic spar varnish. A rail top section, shall be provided on each rack panel over the plywood edge, which has openings approximately two by six inches to facilitate removal of each rack by forklift trucks. The top of rack assemblies, with the upper section of the rail, and the top of the front panel shall be flushed together. The side and rear rack sections and all posts shall be equipped with a quick disconnect permanently attached device to prevent loss and to retain racks to the stake pockets. The rear corners at the top of the racks shall be secured together with a chain with a snap which is attached to the rear corner post (see 4.8.2).

3.4.14.2.1 Racks waterproofness. The racks shall be waterproofed and shall withstand soaking in water without swelling, or otherwise deteriorating, or losing the characteristics of easy installation and removal (see 4.8.4.6).

3.4.14.3 Rack panels. When the racks are installed, the wood panels shall be inside the load space with the upright posts outside, and all corners shall be square. All rack panel fasteners shall be cadmium or zinc-coated steel (see 4.8.2).

3.4.14.4 Stakes. The side and the rear racks shall have stakes that are spaced, 24 $\pm$ 1/16 inch and 23 $\pm$ 1/16 inches apart respectively, between centers. The stakes shall be extruded aluminum (6061-T6, 6063-T5, or 6063-T6) not less than 0.094 inch thick. The aluminum thickness of the "dog ears" on the removable stakes shall be approximately .188 inches for dent resistance. The stake posts of the aluminum extrusions at the lower ends (where they fit into the stake pockets) shall have basic maximum dimensions of 3.375 x 1.500 in. The radii of the posts (interfacing stake pocket corners) shall be 0.375 inches minimum to preclude binding in the stake pocket. Also, any sharp corners must be ground off at 45 degrees (°) while maintaining 0.125 inch wall thickness to preclude binding in the stake pocket corners. The top of each post (fixed and removable) shall have a one inch opening for the installation of 7/8 inch tubular tarp bows described in 3.4.14.6. The removable posts shall have 0.875 inch channel openings to interface with the 5/8 inch RIFS plywood panels. Stakes and panels, including hardware, shall be held together with bolts or other through-type fasteners at least 1/4 inch in diameter (see 4.8.2).

3.4.14.5 Chains. Four chains, two on each side, shall be provided at equidistant points from the top of the rack to the opposite side of the trailer bed floor. The cross chains shall be used for securement to the breakbulk cargo tiedowns on the trailer bed floor and to the top of the

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racks. A 7/16 inch minimum diameter hole for 3/8 inch bolt shall be provided at each top rail section. One end of each chain shall be fitted with grab hooks. The securement system when not in use shall allow for the sliding of cargo without restriction or interference (see 4.8.2).

3.4.14.6 Tarpaulin. When specified (see 6.2), each semitrailer shall be equipped with a one-piece tarpaulin. The tarpaulin shall fit the semitrailer when the side and rear racks are installed. The tarpaulin material shall be vinyl-coated nylon conforming to type II, class 2 of MIL-C-20696. It shall be 12 ft wide by 44 ft long, such that when installed, there will be an approximate 18 inch drop over the side and rear racks and an 18 inch drop over the front panel. The tarpaulin color shall be lusterless forest green and have infrared reflectance properties as specified in MIL-E-52798. All edges of the tarpaulin shall have 3-inch hems with two rows of stitching. There shall be four 10-inch wide reinforcing wear strips of vinyl-coated nylon, as specified above, located longitudinally and transversly on the inside of tarpaulin. All strips shall have two rows of stitching and all stitching shall be two-line, conforming to V-T-285, type I, class I, color olive drap, size 4. Adhesive shall be applied to all seams and liners. Type of adhesive and type of thread sealant shall be approved by the government. Thread sealant shall be applied to stitching. The tarpaulin shall have 54 heavy duty spur type grommets and steel D-rings around the hem to secure the tarp to the trailer through the use of shock straps and tiedown ropes. Grommets and D-rings shall be cadmium or zinc-plated, forest green or black color (see 4.8.1 and 4.8.2).

3.4.14.6.1 Tarpaulin tiedown provisions. The following shall be added to the semitrailer to permit attachment of some tideown ropes, shock straps, safety ropes, and ridge rope (see 4.8.2).

a. Four D-rings shall be welded to the lower front bulkhead assembly. One D-ring shall be welded to the top center of the nose box cap.
b. Rear trailer tiedown rail steel strip 2.00 x 0.25 inches welded across and under rear bumper rail. Strap shall have twelve 0.50 inch diameter holes positioned equidistant. These holes will be used for attaching the shock strap assemblies. Optionally, D-rings may be used.
c. One D-ring shall be welded to the center of the rear tarp tiedown strip. This D-ring and the one at the nose box shall be used to secure the ridge rope.
d. Two D-rings shall be welded, one on each side, to the inside of the outside rails to attach the side safety ropes.

3.4.14.6.2 Tarp. The front of the tarp shall be contour sewn, such that there is no opening at the corners, and it will form-fit the front end of the trailer. The rear end shall have plain corners that will be wrapped around the rear end and secured with heavy duty tarp shock straps. The four corners of the tarp, at the points where the top of the side-racks meet the top of the rear racks and front panel, shall be reinforced inside with 32 ounces (oz) vinyl-coated nylon (10 inches diameter) to reduce stress at these points. A safety tiedown rope shall be provided on each side between bow

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number 5 and 6 to retain the tarp if shock straps should fail or disengage. Five tarp tiedown ropes shall be attached on front of tarp to tie down the tarp to the trailer front rail. A tarp gable ridge rope 55 feet long, .38 nylon, shall be provided to prevent sagging of tarp between bows. The tiedown ropes shall be polypropylene conforming to MIL-R-24049, type II, three strand, twisted, monofilament, 1/4 inch diameter, color black. The nylon gable ridge rope shall be captive sewn into the tarpaulin on the inside top center in such a manner as to allow longitudinal movement of a minimum of 12 inches (see 4.8.1 and 4.8.2).

3.4.14.7 Bows. When specified (see 6.2), each semitrailer shall be equipped with ten interchangeable 0.875 inch tubular bows, 0.083-inch wall thickness, mechanical tubing and steel in accordance with ASTM A513. Tubing shall be hot rolled, pickled and oiled, and electric welded. The ends of the hollow tubing shall have inserted therein up thru the curved portion a 0.625 inch diameter reinforcing bar M1020 steel, in accordance with ASTM A36. The 54 spur grommets shall meet the requirements of MS20230. Each bow shall have a 4 1/2 inch rise over the top of panels and posts with a ridge rope recess in the middle of each bow to retain a tarp gable ridge of .38 inch diameter. Bows shall protrude 4 inches into the top of the removable stakes and have a stop washer attached to maintain bow in that position. The bows shall normally mount into the top of all fixed stake posts on the side racks described in 3.4.14.2 (see 4.8.1 and 4.8.2).

3.4.14.8 Straps. Forty-nine shock straps with steel S-hooks shall be provided to secure the sides and rear of tarp to the trailer side and rear rails. The shock strap assemblies shall attach securely to the grommets and D-rings in the hem of the tarp. One of S-hooks on each shock strap shall be securely attached to the grommet and D-rings to preclude inadvertent loss. The grommet opening shall be a minimum of 9/16 inch diameter. The D-ring shall be a minimum of 5/32 inch diameter steel wire in accordance with MS87006. The rubber straps shall function at a temperature range of +125°F to -40°F. Of the 49 shock strap assemblies, there shall be 44 standard 15 inch rubber straps on the side, three standard 21 inch rubber straps on the rear, and two standard 31 inch rubber straps on the rear corners of the tarpaulin. The rubber shock strap cross section shall be a minimum of 1/4 x 3/4 inches (see 4.8.2).

3.4.15 Stowage. Provisions shall be incorporated into each semitrailer to provide secured storage space for side racks, tarpaulin and bows, ropes, straps, BII, tiedown devices and manifests (see 4.8.2).

3.4.15.1 Rack stowage boxes. Provision shall be incorporated into each vehicle to store the racks. Boxes shall have provisions for securement of doors by staples and swivel snaps and for holding in an open position. Each corner shall have a 3/8 inch diameter drain hole. The design of the doors shall allow for locking the compartment with a padlock and the design of the load bearing surface area shall support the stored racks without bending (see 4.8.2).

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3.4.15.2 Manifest holder. A waterproof manifest holder with a minimum inside dimension of 8 1/2 x 11 x 1 inches shall be attached to the forward edge of the front panel (see 4.8.2).

3.4.15.3 Storage area. The semitrailer shall incorporate a storage box with a minimum of 3.8 cubic feet of area for secured storage of load binders, load securing hardware, and BII normally applicable for the operations intended. The stowage shall be ahead of the axles. Access shall be attainable when the semitrailer is loaded with any container payload. The storage box doors shall have provisions for securement by staples and swivel snaps fastening with padlock and for holding in open position. Drain holes (3/8 inch diameter) shall be located at each corner of the box (see 4.8.2).

3.4.15.4 Tarpaulin and bows stowage. A fully inclosed stowage box shall be provided on the semitrailer to stow the tarpaulin, ropes and straps. This box shall be on the road side, positioned in front of the rack stowage boxes. The box shall have a hinged door with one staple to secure the door closed. In addition, there shall be provisions for locking the stowage compartment with a padlock. A chain shall be welded to the door and a hook welded to the inside of the outside rail to provide provisions for holding the door up and open. Four hanger assemblies, two on each side, shall be provided to store the tarp bows vertically above the rack storage boxes, five bows per side. One staple assembly and chain shall be provided on each side to allow securement of the bows (see 4.8.2).

3.4.16 Deck. The semitrailer deck shall present no restrictions on the sides or end to the opening and closing of the doors of mounted containers or to access by forklifts to tine pockets in pallets and TRICONs located on the semitrailer deck (see 4.8.2).

3.4.17 Kit for yard use. When specified (see 6.2), a kit shall be provided to adapt the semitrailer for use as a yard vehicle. The kit shall be capable of guiding both 20 and 40 ft containers on to the semitrailer bed for rapid loading with a minimum of six inch guidance in each direction. Each vehicle shall have pre-cut slots for the rear guides of the yard kit; however, pre-drilled holes for attaching the eight side and two rear guides shall not be provided. Instruction for drilling such holes shall be provided with the yard kit (see 4.8.2).

3.4.18 Maintenance characteristics. The semitrailer design, together with components and accessories, shall permit ready access to all items requiring periodic service or maintenance in the field. Periodic service and maintenance shall be accomplished with minimum disturbance to other components of the semitrailer (see 4.8.2).

3.4.19 Transportability. The fully loaded semitrailer shall be equipped with lifting and tiedown provisions that allow the semitrailer to be lifted and secured on a railroad flat car, break-bulk ship, or C141 aircraft. The applicable transportability criteria is set forth in MIL-STD-1366. The semitrailer shall be capable of negotiating a 30° ramp (see 4.8.4).

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3.4.19.1 Slinging provision. The semitrailer with rated payload shall be equipped with lift points to permit slinging a fully loaded semitrailer in accordance with type II of MIL-STD-209. In addition, the semitrailer shall be suited for lifting through container locking devices, when secured to a 40 ft ISO container with vertical acceleration of two gravities (g) (see 4.8.4.1).

3.4.19.2 Rail impact. The fully loaded semitrailer shall withstand the impact forces encountered in rail transportation (see 4.8.4.2).

3.4.19.3 Air transport. The semitrailer shall conform to the air transportability requirements of MIL-A-8421. Preparation of the semitrailer for air transport shall be accomplished in 15 minutes total or less by no more than two individuals using only common tools on board (see 4.8.4.3).

3.4.19.4 Vehicle tiedown. The semitrailer shall be equipped with tiedown provisions to permit its transportation, with rated payload, in accordance with type II, class 2 of MIL-STD-209. The fully loaded semitrailer shall be suited for Trailer-On-Flat-Car (TOFC) transportation using any American railroad car designed for this purpose. In addition, the semitrailer shall be provided with tiedown provisions to permit the loading, securement and transport of two each M113, full tracked personnel carrier, chassis series in accordance with type I, class 2 of MIL-STD-209. The tiedown provisions shall not deform when the semitrailer is subjected to rail impact tests (see 4.8.4.4).

3.4.19.5 Container securements and locking devices. The container securement devices shall meet the requirements of FMVSR 393.100 and AAR M943 with rated load in containers and shall withstand load conditions equal to or exceeding the longitudinal container acceleration of 2.5 g relative to the semitrailer. Locking devices shall be able to withstand the loads specified herein when only two of the locks take up the entire rated load. Both the breakbulk cargo and the container securement hardware shall secure the rated cargo weight, under all conditions of transport specified in MIL-STD-209, under TOFC conditions, and in an eight mph hump test in both directions, using a conventional type rail flatcar (see 4.8.4.5).

3.4.20 Breakbulk cargo tiedowns. The semitrailer shall be equipped with ten breakbulk cargo tiedowns, five each per side, in accordance with type II, class 5 of MIL-STD-209. The breakbulk cargo liedowns shall be capable of being used concurrent with the side racks (see 4.8.1 and 4.8.2).

3.4.20.1 Container locks. When the securement system includes container locks, multi-position container locks, they shall be with securing "F" pins. If a retractable type of container locking device is used at all container loading positions, they shall be of identical design, the two front retractable may be a different design. The "F" lock pins and the removable container locks of the container securement system shall be connnected together by a suitable cable assembly. The method of attaching the "F" lock pin to the container locking device shall not fail or separate under normal handling conditions (see 4.8.1 and 4.8.2).

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3.4.20.2 Underslung tiedown rails. The semitrailer shall be equipped with underslung steel tiedown strips, 2 x 5/16 inches, under the stake pockets along the full length of the side rails of the trailer. These strips shall be used for the purpose of expediently tying down breakbulk cargo with available rope and other devices when standard tiedown devices are not available. The design shall be such as to permit the installation of the side racks, specified in 3.4.14.2, and the current bolt and pin assemblies used to retain the racks to the vehicle. The tiedown rail may be broken to be compatible with hardware such as semitrailer sling points and fixed position container locks (see 4.8.2).

3.4.20.3 Missile system securement kits. The semitrailer shall be provided with 14 sets of "chicken track" tiedowns, seven each per side. When specified (see 6.2), the semitrailer shall be provided with a missile system securement kit. The kit when installed shall be capable of tying down nuclear weapons, guided missiles, and large rocket components utilizing 27 web strap assemblies, conforming to Drawing 10900880, rated at 5,000 lb each. Each kit requires a total of 54 tiedown fittings, 27 on each side of the semitrailer (see 4.8.2).

3.4.21 Wheel loading. The three axles of the semitrailer shall be positioned so that the portion of the GCW imposed on the tires, with rated payload evenly distributed over the load space, shall result in equal axle loadings on each of the tractor bogie axles and the semitrailer axles, assuming a tractor bogie weight, without the semitrailer, of 10,000 lb and no weight transfer to the front tractor axle (see 4.8.2).

3.5 Performance.

3.5.1 Automotive. The fully equipped semitrailer shall evidence no part failure or permanent set when towed and operated by a suitable tractor and shall conform to the requirments specified herein during the operational profile specified in table I (see 4.8.5.1).

3.5.2 Tractor compatibility. The semitrailer shall be compatible with the truck-tractors specified in 6.1. The fully loaded semitrailer, when coupled to a truck-tractor, shall be capable of negotiating an abrupt crest on a 12% grade, without interference (see 4.8.5.2).

3.5.3 Turning ability. The semitrailer shall be capable of assuming a 90° angle in both directions to the coupled towing vehicle without cramping, side-slipping, or damage to the semitrailer or the towing vehicle (see 4.8.5.3).

3.5.4 Side slopes. The loaded semitrailer when towed by the truck-tractors specified shall be capable of operating on a 15% side slope without slipping or upsetting with the load center of gravity 51 inches above the deck under all operating conditions (see 4.8.5.4).

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3.5.5 Fording ability. The semitrailer shall ford salt and fresh water to a depth of 18 inches without any special servicing or protection before fording. No evidence of water leakage or moisture penetration into sealed wheel bearings, shall be permitted (see 4.8.5.5).

3.5.6 Tracking ability. The semitrailer shall conform to the tracking requirements of FMCSR 393.70 (see 4.8.5.6).

3.6 Painting. All semitrailers shall be painted to provide a low reflectance CARC surface. The final painted surface shall be three color camouflage pattern applied in two coats at least 1.8 mils thick (total), without sags, runs or thin areas and shall conform to MIL-C-46168 or MIL-C-53039. The front curbside rack (on the inside) paint a "1" in the middle of the upper half of the rack. Then number clockwise around the bed. Paint the numbers so that the black will be used over a green or brown background. Use the camouflage green paint on a black background. The three color pattern segments required shall be as follows (see 4.8.2):

a. 41 ± 5% of the surface area shall be of the color black, conforming to chip 37030 of FED-STD-595.
b. 15 ± 5% of the surface area shall be of the color camouflage brown 383, conforming to chip 30051 of FED-STD-595.
c. 44 ± 5% of the surface area shall be of the color camouflage green 383, conforming to chip 34094 of FED-STD-595.

3.6.1 Cleaning and treating. The semitrailer and components shall be cleaned and treated in accordance with MIL-STD-193 except that all hardware and trim not normally painted shall be treated to provide a matte finish. In addition, contact areas of ferrous sheet metal 0.125 inch thickness and under, which cannot be primed after welding or are not zinc clad steel, shall be cleaned in accordance with TT-C-490 and primed in accordance with MIL-P-46105 (see 4.8.2).

3.7 Rustproofing. The semitrailer shall be rustproofed in accordance with MIL-R-46146 (see 4.8.1).

3.8 Welding. Welding shall not be employed as a repair measure for defective parts. Welding procedures shall be in accordance with class 2 of MIL-STD-1261. Weld penetration shall be such as to provide transfer of maximum design stress through base metals. Fillet welds shall be provided when necessary to reduce stress concentration (see 4.8.1).

a. Weld joint design and construction shall provide adequate material for transferral of load through the joint from one member to another without significant increase in stress.
b. All welders shall be tested and qualified and submit specimens (samples and actual parts) simulating all welding conditions. Specimens shall be available to the Government quality assurance representative for each welder prior to

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performing production welding. The Government quality assurance representative may require requalification of welders, welding equipment operators or equipment to the recorded welding procedure when the quality is suspect based on inspection results.

3.9 Marking. The semitrailer exterior markings shall be placed and sized in accordance with MIL-STD-642. Markings, letters and numbers on the semitrailer shall be lusterless paint conforming to MIL-C-53039 or MIL-C-46168. Black chip 37030 of FED-STD-595 shall be used over a green or brown background. Camouflage green 383, chip 34094 of FED-STD-595, shall be used over black background (see 4.8.2).

3.9.1 Instructional markings. Markings, such as tire pressure, tiedown points, center of gravity (empty), etc. shall be applied to the semitrailer in block letters or numbers from 7/8 to one inch in height (see 4.8.2).

3.9.2 CARC markings. All semitrailers shall have the word "CARC" painted in block letters in a conspicuous location as near to the data plates as practical. Height of block letters shall be in the range specified in 3.9.1 (see 4.8.2).

3.9.3 National symbol. A three inch star shall be applied on the semitrailer in accordance with MIL-STD-642 (see 4.8.2).

3.9.4 Container locks. If multi-position container locks are incorporated, usage locations shall be stenciled on the outboard side of the side sill at each location using one inch, in height, numbers (see 4.8.2).

3.9.5 Registration numbers. Registration numbers shall be placed on both sides and rear of the semitrailer, as near as possible to center line, in uniform gothic four inches high, characters (optional height of three inches). All characters shall be continuous (see 4.8.2).

3.9.6 Tarpaulin. The tarpaulin shall be indelibly marked on the inside in two inch high gothic black lettering with the words "PROPERTY OF THE US ARMY" (see 4.8.2).

3.10 Identification plates. Identification plates shall be in accordance with type I, style I, compositon C of MIL-P-514. Vehicle registration number and manufacturer's serial number shall be placed on a data plate. The tranportation, instruction, safety, warning, caution, air transport, tiedown and notice plates shall be in accordance with type III, composition C of MIL-P-514. The size of the plates shall be within the area of the trailer and shall be legible and readily viewable to service personnel standing on level ground; plates shall be attached to the web exterior of the main frame rail on the curb side (see 4.8.1 and 4.8.2).

3.10.1 Vehicle shipping data plate. A shipping data plate shall be included and shall show the silhouette of the vehicle in transport position indicating the location and capacity of the tiedown attachments, and supplementary points of tiedown. A data plate showing slinging through a 40

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ft container shall be provided. The plate shall warn against slinging through any other type container or combination of containers. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location. Data plate shall be as specified in 3.10 and shall show the location and capacity of the lift points, vehicle tiedown points and breakbulk cargo tiedown points. The silhouette of the semitrailer is to show dimensionally the longitudinal location of the vehicle center of gravity (with and without rated payload, 34 tons evenly distributed), location of the landing gear, axles (including axle spacing), deck height, minimum ground clearance, height at upper fifth wheel plate, overall height (with and without payload), overall length and overall width. A chart shall be provided showing axle, kingpin, and landing gear loads for both curb weight and gross vehicle weight (GVW) (see 4.8.2).

3.10.2 Lubrication plate. The semitailer shall be provided with a lubrication plate conforming to SAE J753 installed in a place adjacent to the identification plate. The plate shall specify lubricants and their temperature range (see 4.8.1 and 4.8.2).

3.11 Lubrication. Lubrication means shall be provided for all parts of the semitrailer normally requiring lubrication. Grease lubrication fittings shall conform to SAE J534. When the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used (see 4.8.1 and 4.8.2).

3.12 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for immediate operational use. The servicing and adjusting shall include at least the following: Inflation of all tires, adjustment of brakes, proper functioning of all lighting, and complete lubrication with grades of lubricants recommended for ambient temperatures at the delivery point (see 4.8.2).

3.13 Workmanship. Workmanship shall be of a quality which will assure a product free of rust, scratches, chips, sharp edges, loose or defective connectors, faulty welding, or other defects which may affect serviceability or performance (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operation to determine that practices, methods, and procedures of the written quality assurance system plan are being properly applied, and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedure, or instances of poor practices which might have an adverse effect upon quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until the conformance of the product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. When a part or component is specified to conform to a specification having a Qualified Products List (QPL), the contractor shall make available to the Government documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity.

4.1.5 Quality assurance provisions. In the conduct of inspection, the contractor shall adhere to Quality Assurance Provisions (QAP) and General Quality Assurance Provision (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.1.6 Certification. Where certification (see 4.8.1) is required to verify material or component conformance to a specification, the contractor shall furnish such certification along with documented test results and performance and analytical data, as applicable.

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4.2 Classification of inspection:

a. First article inspection (see 4.3).
   1. First production vehicle inspection (FPVI) (see 4.3.1).
   2. Initial production test (IPT) (see 4.3.2).

b. Quality conformance inspections (QCI) (see 4.4).
c. Control tests (see 4.5).
d. Comparison tests (see 4.6).

4.3 First article inspection. First article inspection shall be performed on the first production semitrailer and on three additional semitrailers, randomly selected by the Government, which shall be designated as initial production vehicles. Approval of the first article samples shall not relieve the contractor of the obligation to supply semitrailers that are fully representative of those inspected as a first article sample. Any change or deviation of production semitrailers from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 FPVI.

4.3.1.1 In-process inspection. The first production semitrailers shall be inspected during fabrication to determine conformance of materials and workmanship to specified requirements. These inspections shall preceed application of primer and paint. Processing and welding procedures, the quality system, inspection records, calibration procedures, radiographic procedures welder certifications shall be reviewed and evaluated.

4.3.1.2 Completed vehicle inspections.

4.3.1.2.1 Contractor inspection. The contractor shall subject the vehicle to a road test and the FPVI specified in table III. The road test shall be conducted with actual or simulated payload on a smooth, relatively level hard-surfaced road for a distance of not less than 50 mi.

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TABLE III. Classification of inspections.

Title	Requirements	Inspection	First article		1/ QCI	1/ Control	2/ Comparison
			1/ FPVI	2/ IPT			
Materials and construction	3.3 thru 3.4, 3.4.3, thru 3.4.5, 3.4.7, 3.4.7.1, 3.4.8, 3.4.12.1, 3.4.12.2, 3.4.14.6, 3.4.14.7, 3.4.14.8, 3.4.20, 3.4.20.1, 3.7, 3.8, 3.10, 3.10.2, and 3.11	4.8.1	X	X	X		
Defects (see 4.4 and table IV)	3.4 thru 3.4.2, 3.4.4, 3.4.6 thru 3.4.12.1, 3.4.12.3 thru 3.4.14.2, 3.4.14.3 thru 3.4.18, 3.4.20 thru 3.4.21, 3.6, 3.6.1, and 3.9 thru 3.13	4.8.2	X	X	X	X	X
Life cycle operations	3.2	4.8.3	X	X			
Payload and distribution	3.2.1	4.8.3.1	X	X			X
Tracked vehicle payload	3.2.1.1	4.8.3.1	X	X			X
Environmental	3.2.2	4.8.3.2	X	X			X
Reliability	3.2.3	4.8.3.3	X	X			X
Maintainability	3.2.4	4.8.3.4	X	X			X

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TABLE III. Classification of inspections - Continued.

Title	Requirements	Inspection	First article 1/ FPVI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Transportability	3.4.19	4.8.4	X	X			
Slinging provision	3.4.19.1	4.8.4.1	X	X	X	X	X
Rail impact	3.4.19.2	4.8.4.2	X	X			
Air transport	3.4.19.3	4.8.4.3	X	X			
Vehicle tiedown	3.4.19.4	4.8.4.4	X	X			
Container securements and locking devices	3.4.19.5	4.8.4.5	X	X			
Racks waterproofness	3.4.14.2.1	4.8.4.6	X	X			
Performance tests	3.5	4.8.5	X	X			
Automotive	3.5.1	4.8.5.1	X	X	X	X	X
Tractor compatibility	3.5.2	4.8.5.2	X	X	X	X	X
Turning ability	3.5.3	4.8.5.3	X	X	X	X	X
Side slopes	3.5.4	4.8.5.4	X	X		X	X
Fording ability	3.5.5	4.8.5.5	X	X			X
Tracking ability	3.5.6	4.8.5.6	X	X	X	X	X

1/ At place of manufacture
2/ At Government proving ground

4.3.1.2.2 Preliminary inspection. The responsible Government inspection element shall conduct the FPVI specified in table III to the extent that acceptability of the semitrailer is verified.

4.3.1.2.3 Provisional inspection. The Government procuring activity shall perform the FPVI specified in table III to the degree required to justify approval of the semitrailer.

4.3.1.2.4 Vehicle disposition. After completion of inspection, the first production semitrailer shall remain at the manufacturing facility, as a production sample, and be the last vehicle of its type shipped on the contract. The semitrailer may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the semitrailer during this period in accordance with applicable documents for care and preservation while in storage. All configuration changes taking

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place after the first production inspection, shall be made to the first production semitrailer so that the semitrailer will be representative of the current configuration throughout the life of the contract. No configuration changes may be implemented on production vehicles after Government conditional acceptance of the first production vehicle (applicable vehicle type) without written authorization from the Government.

4.3.1.3 Shakedown test. Following first vehicle inspection and initial road test, the semitrailer shall be subjected to a 2000 mi shakedown test by the contractor in accordance with table I except that the mileages shall be 1000 mi each on hard-surfaced and secondary roads with 900 mi each at maximum payload and 100 mi each without payload. Test results shall be submitted by the contractor within 15 days of the test and shall include the demonstration performed, deficiencies encountered and action taken to correct the deficiencies.

4.3.2 IPT.

4.3.2.1 Contractor inspection. The contractor shall inspect the initial production test semitrailers in accordance with 4.4 (see 6.2).

4.3.2.2 Government inspection. The Government shall perform the IPT specified in table III, a 6,000 mi test according to the operational profile specified in table I, and a 5450 mi test as specified in 4.8.3.1. IPT shall require a maximum of 180 days per trailer to complete test.

4.3.3 Deficiencies. Unresolved failures or deficiencies during, or as a result of, first article inspection shall be cause for rejection of the vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the IPT shall be evidence that all vehicles already accepted prior to completion of the IPT are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnishd by the contractor. Such deficiencies on all vehicles shall be corrected by the contractor.

4.4 QCI. QCI shall be performed in the following sequence and utilizing an approved Final Inspection Record (FIR).

a. Examination for the defects specified in table III and classified in table IV.
b. A road test of not less than 15 mi in accordance with 4.4.4.
c. QCI specified in table III.

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TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly incomplete (see 3.4)	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.4 and 3.4.1).	SIE 1/
103	Improper welding affecting performance (see 3.8).	Visual
104	Improper markings (see 3.9).	Visual
105	Improper data plates (see 3.10).	Visual
106	Faulty workmanship affecting performance (see 3.13).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.4 and 3.4.1).	SIE
202	Improper paint, cleaning, treating, or rustproofing (see 3.6 and 3.7).	Visual
203	Improper markings (see 3.9).	Visual
204	Improper plates and data instructions (see 3.10).	Visual
205	Faulty workmanship affecting appearance (see 3.13).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.1 One hundred percent. Each vehicle shall be subjected to the QCI specified in 4.4.

4.4.2 Acceptable quality level. When authorized by the Government contracting authority, 100% QCI (see 4.4.2.1) shall be waived. Thereafter, samples shall be selected in accordance with 4.4.2.1.2 and shall be inspected to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.4.2.1 Sampling.

4.4.2.1.1 Lot formation. Unless otherwise specified (see 6.2), an inspection lot shall consist of all the semitrailers produced in one month submitted at one time for acceptance.

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4.4.2.1.2 Sampling for examination. When sampling is authorized for QCI (see 4.4.2) samples shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall 100% examine at least 50 consecutively produced semitrailers. The examination shall establish that the process average percent defective, as defined in MIL-STD-105 and in 6.4, is not greater than the specified AQL.

4.4.3 Defects. The following rules shall be applied to the counting of defects.

4.4.3.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a 30-day period, will be added to the minor defects classification with no increase in AQLs.

4.4.3.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

4.4.3.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

4.4.4 Road test. Each semitrailer, without payload, shall be operated for distance of not less than 15 mi by the contractor at speeds up to maximum. The semitrailer shall be completely assembled and serviced, less on vehicle equipment (OVE), prior to road test. Performance of semitrailers shall be demonstrated on smooth, relatively level hard-surfaced roads. After completion of 15 mi road test, the semitrailer shall be checked for lubrications leakage and other deficiencies. Semitrailers shall then be run for an additional five mi to recheck the corrections made as a result of the initial 15 mi road test.

4.5 Control test. Control tests shall be conducted on one semitrailer per month or on one from each lot of 50 semitrailers consecutively preproduced, except that not more than two tests nor less than one test shall be performed in a 30-day period. The vehicles shall be road tested in accordance with the road test specified in 4.5.1 and shall subsequently be subjected to the control tests specified in table III.

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4.5.1 Fifty mile test. Semitrailers shall be equipped with either actual or simulated payload and operated for a distance of not less than 50 mi. All BII and OVE that is packed with the semitrailer shall be installed and inspected for fit, count and condition. The test course used for this test shall be a smooth, relatively level, hard-surfaced road. After the test the semitrailer shall be examined for leaks, damage, cracks and permanent set.

4.6 Comparison test. The Government may randomly select vehicles at anytime during the production contract period and subject these vehicles to the comparison tests specified in table III as well as a 2,500 mi test in accordance with the operation profile specified in table I. Tests shall be conducted by the Government at a site it shall select. Vehicles selected for comparison tests shall not have been previously tested according to 4.5, but shall have passed inspection according to 4.4.

4.7 Failure. Failure of any vehicle to pass any of the specified inspections or failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.3 thru 3.4, 3.4.3 thru 3.4.5, 3.4.7, 3.4.7.1, 3.4.8, 3.4.12.1, 3.4.12.2, 3.4.14.6, 3.4.14.7, 3.4.14.8, 3.4.20, 3.4.20.1, 3.7, 3.8, 3.10, 3.10.2, and 3.11 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.4 thru 3.4.2, 3.4.4, 3.4.6 thru 3.4.12.1, 3.4.12.3 thru 3.4.14.2, 3.4.14.3 thru 3.4.18, 3.4.20 thru 3.4.21, 3.6, 3.6.1, and 3.9 thru 3.13 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Life cycle operations tests. To determine conformance to 3.2, semitrailers shall be subjected to the reliability and maintainability tests specified in 4.8.3.3 and 4.8.3.4 respectively.

4.8.3.1 Payload and distribution. To determine conformance to 3.2.1 and 3.2.1.1, the semitrailer shall be tested for the capability of transporting the loads specified of breakbulk cargo, any combination of containers as specified in table II, and two M113 tracked vehicle series as specified. Testing shall be for a 5450 mi test, 2750 mi on hard surfaced roads at 55 mph and 2700 mi on improved secondary roads at 30 mph. The semitrailer shall then be checked for any permanent deformation as a result of this test.

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4.8.3.2 Environmental conditions. To determine conformance to 3.2.2, the semitrailer, with all its components, shall be operated and stored under the weather and altitude conditions specified without any degradation or malfunctions.

4.8.3.3 Reliability. To determine conformance to 3.2.3, the semitrailer reliability shall be verified during a 6,000 MMBF, with a confidence level above a point estimate, without any extensive damages or performance degradation that cannot be corrected by the crew in the time limit specified. The reliability of the securement system shall be demonstrated during the test by cycling one 20-ft container on and off for 100 cycles.

4.8.3.4 Maintainability. To determine conformance to 3.2.4, the semitrailer shall demonstrate the maintainability characteristics during the 6,000 mi test using the specified manuals. Total maintenance time and MR shall be verified to conform to the values specified.

4.8.4 Transportability tests.

4.8.4.1 Slinging provision. To determine conformance to 3.4.19.1, the semitrailer shall be tested for slinging provisions and for lifting containers with the exterior at specified vertical acceleration in accordance with the test requirements of MIL-STD-209.

4.8.4.2 Rail impact. To determine conformance to 3.4.19.2, a fully loaded semitrailer shall be tested in accordance with MIL-STD-810, method 516, procedure VIII using an APR flat car designed for this purpose.

4.8.4.3 Air transport. To determine conformance to 3.4.19.3, the semitrailer shall be tested for internal air transportability by a C141 aircraft. Air transportability shall be tested in accordance with MIL-A-8421 and MIL-STD-1366. Preparation time for air transport shall be taken and shall be as specified.

4.8.4.4 Vehicle tiedown. To determine conformance to 3.4.19.4, the semitrailer shall be tested for tiedown hardware in accordance with MIL-STD-209 as specified and tested in accordance with 5450-mi test specified in 4.8.3.1 when loaded with two M113 tracked vehicles.

4.8.4.5 Container securement and locking devices. To determine conformance to 3.4.19.5, fully loaded containers shall be secured to the semitrailer and an acceleration is applied as specified. All securement devices shall then be checked and shall meet the specified requirements. Subsequently the fully loaded containers shall be locked to the semitrailer with two of the locks and undergo the specified operation conditions. After the operation the locks shall be checked for any permanent deformation or damage.

4.8.4.6 Racks waterproofness. To determine conformance to 3.4.14.2.1, the racks shall be soaked in water for 48 hours and then checked for any size deformation that would complicate installation and removal.

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4.2 Classification of inspection:

a. First article inspection (see 4.3).
   1. First production vehicle inspection (FPVI) (see 4.3.1).
   2. Initial production test (IPT) (see 4.3.2).

b. Quality conformance inspections (QCI) (see 4.4).
c. Control tests (see 4.5).
d. Comparison tests (see 4.6).

4.3 First article inspection. First article inspection shall be performed on the first production semitrailer and on three additional semitrailers, randomly selected by the Government, which shall be designated as initial production vehicles. Approval of the first article samples shall not relieve the contractor of the obligation to supply semitrailers that are fully representative of those inspected as a first article sample. Any change or deviation of production semitrailers from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 FPVI.

4.3.1.1 In-process inspection. The first production semitrailers shall be inspected during fabrication to determine conformance of materials and workmanship to specified requirements. These inspections shall preceed application of primer and paint. Processing and welding procedures, the quality system, inspection records, calibration procedures, radiographic procedures welder certifications shall be reviewed and evaluated.

4.3.1.2 Completed vehicle inspections.

4.3.1.2.1 Contractor inspection. The contractor shall subject the vehicle to a road test and the FPVI specified in table III. The road test shall be conducted with actual or simulated payload on a smooth, relatively level hard-surfaced road for a distance of not less than 50 mi.

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TABLE III. Classification of inspections.

Title	Requirements	Inspection	First article		1/ QCI	1/ Control	2/ Comparison
			1/ FPVI	2/ IPT			
Materials and construction	3.3 thru 3.4, 3.4.3, thru 3.4.5, 3.4.7, 3.4.7.1, 3.4.8, 3.4.12.1, 3.4.12.2, 3.4.14.6, 3.4.14.7, 3.4.14.8, 3.4.20, 3.4.20.1, 3.7, 3.8, 3.10, 3.10.2, and 3.11	4.8.1	X	X	X		
Defects (see 4.4 and table IV)	3.4 thru 3.4.2, 3.4.4, 3.4.6 thru 3.4.12.1, 3.4.12.3 thru 3.4.14.2, 3.4.14.3 thru 3.4.18, 3.4.20 thru 3.4.21, 3.6, 3.6.1, and 3.9 thru 3.13	4.8.2	X	X	X	X	X
Life cycle operations	3.2	4.8.3	X	X			
Payload and distribution	3.2.1	4.8.3.1	X	X			X
Tracked vehicle payload	3.2.1.1	4.8.3.1	X	X			X
Environmental	3.2.2	4.8.3.2	X	X			X
Reliability	3.2.3	4.8.3.3	X	X			X
Maintainability	3.2.4	4.8.3.4	X	X			X

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TABLE III. Classification of inspections - Continued.

Title	Requirements	Inspection	First article 1/ FPVI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Transportability	3.4.19	4.8.4	X	X			
Slinging provision	3.4.19.1	4.8.4.1	X	X	X	X	X
Rail impact	3.4.19.2	4.8.4.2	X	X			
Air transport	3.4.19.3	4.8.4.3	X	X			
Vehicle tiedown	3.4.19.4	4.8.4.4	X	X			
Container securements and locking devices	3.4.19.5	4.8.4.5	X	X			
Racks water-proofness	3.4.14.2.1	4.8.4.6	X	X			
Performance tests	3.5	4.8.5	X	X			
Automotive	3.5.1	4.8.5.1	X	X	X	X	X
Tractor compatibility	3.5.2	4.8.5.2	X	X	X	X	X
Turning ability	3.5.3	4.8.5.3	X	X	X	X	X
Side slopes	3.5.4	4.8.5.4	X	X		X	X
Fording ability	3.5.5	4.8.5.5	X	X			X
Tracking ability	3.5.6	4.8.5.6	X	X	X	X	X

1/ At place of manufacture
2/ At Government proving ground

4.3.1.2.2 Preliminary inspection. The responsible Government inspection element shall conduct the FPVI specified in table III to the extent that acceptability of the semitrailer is verified.

4.3.1.2.3 Provisional inspection. The Government procuring activity shall perform the FPVI specified in table III to the degree required to justify approval of the semitrailer.

4.3.1.2.4 Vehicle disposition. After completion of inspection, the first production semitrailer shall remain at the manufacturing facility, as a production sample, and be the last vehicle of its type shipped on the contract. The semitrailer may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the semitrailer during this period in accordance with applicable documents for care and preservation while in storage. All configuration changes taking

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place after the first production inspection, shall be made to the first production semitrailer so that the semitrailer will be representative of the current configuration throughout the life of the contract. No configuration changes may be implemented on production vehicles after Government conditional acceptance of the first production vehicle (applicable vehicle type) without written authorization from the Government.

4.3.1.3 Shakedown test. Following first vehicle inspection and initial road test, the semitrailer shall be subjected to a 2000 mi shakedown test by the contractor in accordance with table I except that the mileages shall be 1000 mi each on hard-surfaced and secondary roads with 900 mi each at maximum payload and 100 mi each without payload. Test results shall be submitted by the contractor within 15 days of the test and shall include the demonstration performed, deficiencies encountered and action taken to correct the deficiencies.

4.3.2 IPT.

4.3.2.1 Contractor inspection. The contractor shall inspect the initial production test semitrailers in accordance with 4.4 (see 6.2).

4.3.2.2 Government inspection. The Government shall perform the IPT specified in table III, a 6,000 mi test according to the operational profile specified in table I, and a 5450 mi test as specified in 4.8.3.1. IPT shall require a maximum of 180 days per trailer to complete test.

4.3.3 Deficiencies. Unresolved failures or deficiencies during, or as a result of, first article inspection shall be cause for rejection of the vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the IPT shall be evidence that all vehicles already accepted prior to completion of the IPT are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnishd by the contractor. Such deficiencies on all vehicles shall be corrected by the contractor.

4.4 QCI. QCI shall be performed in the following sequence and utilizing an approved Final Inspection Record (FIR).

a. Examination for the defects specified in table III and classified in table IV.
b. A road test of not less than 15 mi in accordance with 4.4.4.
c. QCI specified in table III.

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TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly incomplete (see 3.4)	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.4 and 3.4.1).	SIE 1/
103	Improper welding affecting performance (see 3.8).	Visual
104	Improper markings (see 3.9).	Visual
105	Improper data plates (see 3.10).	Visual
106	Faulty workmanship affecting performance (see 3.13).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.4 and 3.4.1).	SIE
202	Improper paint, cleaning, treating, or rustproofing (see 3.6 and 3.7).	Visual
203	Improper markings (see 3.9).	Visual
204	Improper plates and data instructions (see 3.10).	Visual
205	Faulty workmanship affecting appearance (see 3.13).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.1 One hundred percent. Each vehicle shall be subjected to the QCI specified in 4.4.

4.4.2 Acceptable quality level. When authorized by the Government contracting authority, 100% QCI (see 4.4.2.1) shall be waived. Thereafter, samples shall be selected in accordance with 4.4.2.1.2 and shall be inspected to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.4.2.1 Sampling.

4.4.2.1.1 Lot formation. Unless otherwise specified (see 6.2), an inspection lot shall consist of all the semitrailers produced in one month submitted at one time for acceptance.

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4.4.2.1.2 Sampling for examination. When sampling is authorized for QCI (see 4.4.2) samples shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall 100% examine at least 50 consecutively produced semitrailers. The examination shall establish that the process average percent defective, as defined in MIL-STD-105 and in 6.4, is not greater than the specified AQL.

4.4.3 Defects. The following rules shall be applied to the counting of defects.

4.4.3.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a 30-day period, will be added to the minor defects classification with no increase in AQLs.

4.4.3.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

4.4.3.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

4.4.4 Road test. Each semitrailer, without payload, shall be operated for distance of not less than 15 mi by the contractor at speeds up to maximum. The semitrailer shall be completely assembled and serviced, less on vehicle equipment (OVE), prior to road test. Performance of semitrailers shall be demonstrated on smooth, relatively level hard-surfaced roads. After completion of 15 mi road test, the semitrailer shall be checked for lubrications leakage and other deficiencies. Semitrailers shall then be run for an additional five mi to recheck the corrections made as a result of the initial 15 mi road test.

4.5 Control test. Control tests shall be conducted on one semitrailer per month or on one from each lot of 50 semitrailers consecutively preproduced, except that not more than two tests nor less than one test shall be performed in a 30-day period. The vehicles shall be road tested in accordance with the road test specified in 4.5.1 and shall subsequently be subjected to the control tests specified in table III.

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4.5.1 Fifty mile test. Semitrailers shall be equipped with either actual or simulated payload and operated for a distance of not less than 50 mi. All BII and OVE that is packed with the semitrailer shall be installed and inspected for fit, count and condition. The test course used for this test shall be a smooth, relatively level, hard-surfaced road. After the test the semitrailer shall be examined for leaks, damage, cracks and permanent set.

4.6 Comparison test. The Government may randomly select vehicles at anytime during the production contract period and subject these vehicles to the comparison tests specified in table III as well as a 2,500 mi test in accordance with the operation profile specified in table I. Tests shall be conducted by the Government at a site it shall select. Vehicles selected for comparison tests shall not have been previously tested according to 4.5, but shall have passed inspection according to 4.4.

4.7 Failure. Failure of any vehicle to pass any of the specified inspections or failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.3 thru 3.4, 3.4.3 thru 3.4.5, 3.4.7, 3.4.7.1, 3.4.8, 3.4.12.1, 3.4.12.2, 3.4.14.6, 3.4.14.7, 3.4.14.8, 3.4.20, 3.4.20.1, 3.7, 3.8, 3.10, 3.10.2, and 3.11 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.4 thru 3.4.2, 3.4.4, 3.4.6 thru 3.4.12.1, 3.4.12.3 thru 3.4.14.2, 3.4.14.3 thru 3.4.18, 3.4.20 thru 3.4.21, 3.6, 3.6.1, and 3.9 thru 3.13 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Life cycle operations tests. To determine conformance to 3.2, semitrailers shall be subjected to the reliability and maintainability tests specified in 4.8.3.3 and 4.8.3.4 respectively.

4.8.3.1 Payload and distribution. To determine conformance to 3.2.1 and 3.2.1.1, the semitrailer shall be tested for the capability of transporting the loads specified of breakbulk cargo, any combination of containers as specified in table II, and two M113 tracked vehicle series as specified. Testing shall be for a 5450 mi test, 2750 mi on hard surfaced roads at 55 mph and 2700 mi on improved secondary roads at 30 mph. The semitrailer shall then be checked for any permanent deformation as a result of this test.

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4.8.3.2 Environmental conditions. To determine conformance to 3.2.2, the semitrailer, with all its components, shall be operated and stored under the weather and altitude conditions specified without any degradation or malfunctions.

4.8.3.3 Reliability. To determine conformance to 3.2.3, the semitrailer reliability shall be verified during a 6,000 MMBF, with a confidence level above a point estimate, without any extensive damages or performance degradation that cannot be corrected by the crew in the time limit specified. The reliability of the securement system shall be demonstrated during the test by cycling one 20-ft container on and off for 100 cycles.

4.8.3.4 Maintainability. To determine conformance to 3.2.4, the semitrailer shall demonstrate the maintainability characteristics during the 6,000 mi test using the specified manuals. Total maintenance time and MR shall be verified to conform to the values specified.

4.8.4 Transportability tests.

4.8.4.1 Slinging provision. To determine conformance to 3.4.19.1, the semitrailer shall be tested for slinging provisions and for lifting containers with the exterior at specified vertical acceleration in accordance with the test requirements of MIL-STD-209.

4.8.4.2 Rail impact. To determine conformance to 3.4.19.2, a fully loaded semitrailer shall be tested in accordance with MIL-STD-810, method 516, procedure VIII using an APR flat car designed for this purpose.

4.8.4.3 Air transport. To determine conformance to 3.4.19.3, the semitrailer shall be tested for internal air transportability by a C141 aircraft. Air transportability shall be tested in accordance with MIL-A-8421 and MIL-STD-1366. Preparation time for air transport shall be taken and shall be as specified.

4.8.4.4 Vehicle tiedown. To determine conformance to 3.4.19.4, the semitrailer shall be tested for tiedown hardware in accordance with MIL-STD-209 as specified and tested in accordance with 5450-mi test specified in 4.8.3.1 when loaded with two M113 tracked vehicles.

4.8.4.5 Container securement and locking devices. To determine conformance to 3.4.19.5, fully loaded containers shall be secured to the semitrailer and an acceleration is applied as specified. All securement devices shall then be checked and shall meet the specified requirements. Subsequently the fully loaded containers shall be locked to the semitrailer with two of the locks and undergo the specified operation conditions. After the operation the locks shall be checked for any permanent deformation or damage.

4.8.4.6 Racks waterproofness. To determine conformance to 3.4.14.2.1, the racks shall be soaked in water for 48 hours and then checked for any size deformation that would complicate installation and removal.

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4.8.5 Performance tests.

4.8.5.1 Automotive. To determine conformance to 3.5.1, all performance characteristics shall be verified with a fully loaded semitrailer, with the loads specified, over hard-surfaced, and improved secondary roads when coupled to the appropriate tractor.

4.8.5.2 Tractor compatibility. To determine conformance to 3.5.2, the semitrailer with payloads and coupled to a suitable truck-tractor shall be operated on a grade as specified and checked for interference with any part of the tractor during and after the test.

4.8.5.3 Turning ability. To determine conformance to 3.5.3, the tractor-semtrailer combination shall be operated as specified and then subjected to a 90° turning angle in both directions. The semitrailer shall be examined for cramping, side slipping, damages or interference with the tractor during and after the test.

4.8.5.4 Side slopes. To determine conformance to 3.5.4, the semitrailer with payloads and coupled to the towing tractor shall be operated on a side slope as specified and shall then be examined for any signs of slipping or upsetting the center of gravity above the deck.

4.8.5.5 Fording ability. To determine conformance to 3.5.5, the semitrailer, without any special servicing or preparation, shall be forded in fresh and salt water, as applicable. Subsequently all seals, air lines, fittings and electrical components shall be examined for contamination.

4.8.5.6 Tracking ability. To determine conformance to 3.5.6, the semitrailer shall be towed over relatively level, smooth, improved and prepared hard-surfaced roads at specified speeds. The semitrailer shall be examined and shall track the towing semitrailer without shifting or swerving from each side of path of the towing semitrailer when it is moving in a straight course.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with MIL-V-62038 and the applicable packaging requirements as specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. The semitrailer (M872A4) covered by this specification is intended for use by the Armed services for transporting containerized or breakbulk cargo of the configurations and weights specified in 3.2.1 in tactical military operations. It can also haul the M113 series of tracked vehicles, with proper securement and chocking, in addition to retractable and removable fittings to engage and secure standard ANSI/ISO containers. The M872A4 can be towed by the military M915 and M915A1, commercial 6 x 4 truck-tractors, and the military tactical five ton truck-tractors M931 and M932.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article inspection is not required (see 3.1).
c. Tarpaulin and bows, if required (see 3.4.14.6 and 3.4.14.7).
d. Kit for yard use, if required (see 3.4.17).
e. Missile securement system kit, if required (see 3.4.20.3).
f. If responsibility for inspections shall be other than as specified (see 4.1).
g. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
h. The number of initial production tests required (see 4.3.2.1).
i. If lot formation shall be other than as specified (see 4.4.2.1.1).
j. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Primary roads. Primary roads are defined as paved roads.

6.3.2 Improved secondary roads. Unpaved roads that have been prepared by grading, filling and compacting and are periodically maintained by filling holes, grading and rolling.

6.3.3 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.4).

6.3.4 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or

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industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.3.5 Contractor, manufacturer. The term "contractor", as used in this specification, is defined as the organization having a direct contract with one Government activity. The term "manufacturer" is defined as the organization actually performing the operations covered by this specification. The contractor may or may not be the manufacturer.

6.3.6 Major and minor defects. The Government inspector will verify that a thorough inspection of each semitrailer is performed by the contractor not only for the listed characteristics, but also for any departures from good workmanship. The Government inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the semitrailer. The defects noted on Deficiency Sheet should contain sufficient description to enable the government inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the semitrailer specifications with definitions contained in MIL-STD-105. Corrective action shall be taken for occuring deficiencies.

6.4 Process average. Sampling may be initiated if the process average value for the first 20 semitrailers inspected, is less than the AQL specified in the classification of defects for major and minor defects. Process average will be computed as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100% inspection shall be performed and continued until such time that the process average for 20 consecutive semitrailers is less than the specified AQL.

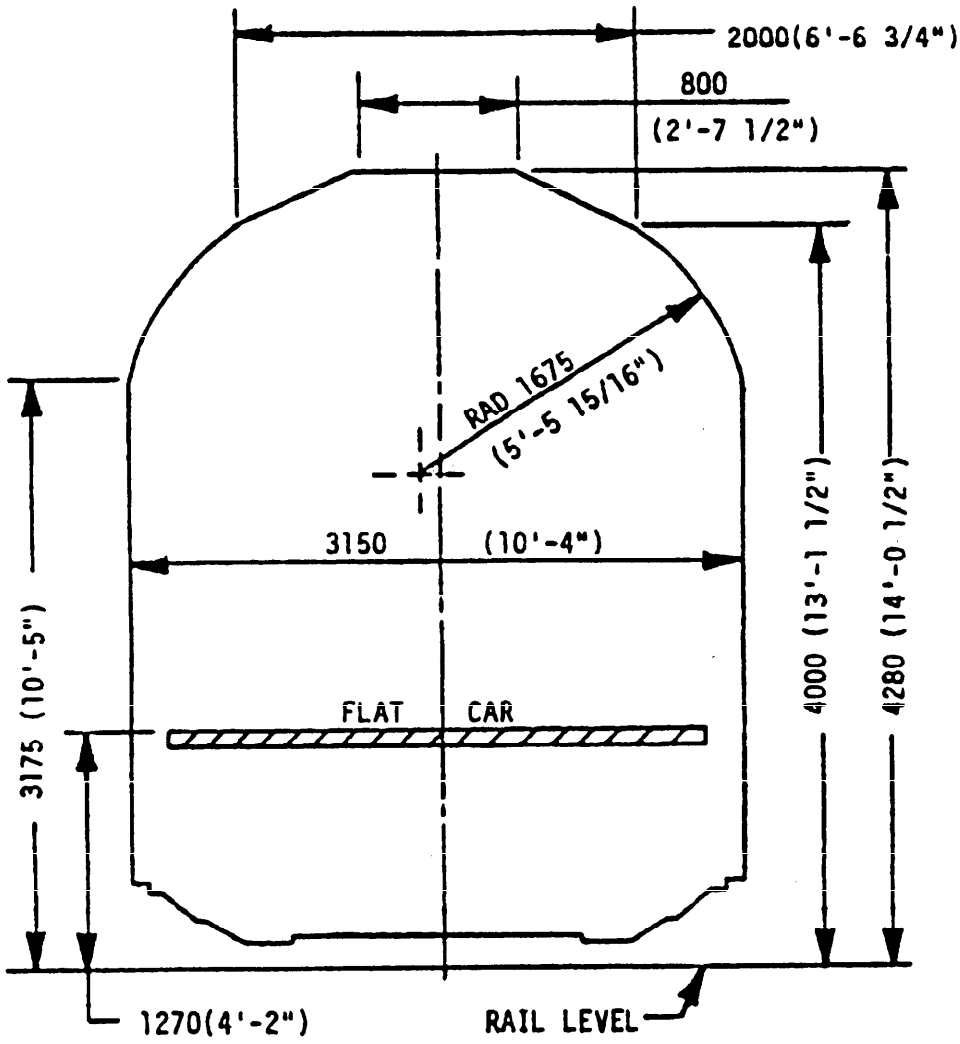
6.5 Rail shipment. The loading of vehicles on open-top cars shall be in accordance with the applicable requirements of the Association of American Railroads, "Rules Governing the Loading of Commodities on Open-Top Cars".

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Subject term (key word) listing.

Breakbulk and container transporter
Flatbed semitrailer, 34 ton
Semitrailer, platform, M872A4
Three-axle, dual purpose semitrailer

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NOTE:
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FIGURE 1. Passe-Partout International (PPI) guage.

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Custodian:
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Preparing activity:
Army - AT

(Project 2330-A955)

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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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